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Bearing fault diagnosis method based on nonlinear features of threshold-free recurrence plot and parallel scale fusion models

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ABSTRACT

In order to ensure the safe and stable operation of mechanical equipment, reduce the risk of failure, and improve the efficiency of operation and maintenance, mechanical fault diagnosis technology has become a key research direction in the field of modern intelligent manufacturing. In this study, based on the previous experimental research, the dynamic characteristics and nonlinear features embedded in the signal are deeply explored. First, the optimal number of layers of wavelet packet decomposition is adaptively determined according to the decreasing information entropy criterion(EDWPD),and the original vibration signal is decomposed into multiple subband signals to realize the expression of information on different frequency scales. Subsequently, the corresponding threshold-free recurrence plot(TFRP)matrices are constructed for each subband signal to describe the nonlinear evolution behavior of the signal in phase space. Based on this, the eigenmode features are extracted, and the eigenvalue decomposition(EVD)is performed on the TFRP matrix to extract the eigenvalues as discriminative features characterizing the eigendynamic modes of the signals, thus constituting the final eigenvectors, which are then inputted into the PSF-CNN-BiLSTM model constructed in this paper for training and testing In this paper, experiments are carried out on the CWRU dataset and the Paderborn dataset, and the results show that the method possesses not only good nonlinear modeling capability but also achieves effective compression and representation of high-dimensional features while maintaining effective discriminative performance and exhibits strong robustness and classification performance. The proposed fault diagnosis method achieves high accuracy on both datasets.

1. Introduction

In the field of reliability engineering for complex systems, the inevitability of system failure stems from multidimensional uncertainty factors. This failure mechanism is particularly significant in complex systems with nonlinear interactions, time-varying characteristics, and high-dimensional coupling relationships. The occurrence of system failure will not only destroy its functional integrity but also may cause significant economic and social losses. Bearings play a critical role in mechanical systems as they endure significant cyclic stresses, making them highly susceptible to wear and failure. Their operational stability is crucial for ensuring the overall reliability of the machinery. [1] Rolling bearings serve as critical elements in rotating machinery, and their operational

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condition plays a decisive role in the equipment's overall dependability. [2] Studies have shown that about 30 % of rotating machinery failures originate from rolling bearing defect [3,4], which highlights the critical engineering value of bearing fault diagnosis. Among many monitoring means, vibration signal analysis has become the preferred method for bearing condition monitoring due to its high sensitivity to early failure [5,6]. However, the actual collected bearing vibration signals often show significant non-stationary characteristics, which brings challenges to fault characterization. For this reason, time–frequency analysis techniques have emerged, Typical transformation approaches encompass STFT-based time–frequency analysis, EMD-based adaptive decomposition, and continuous wavelet-based time-scale representation [7,8], which lays an essential foundation for subsequent fault feature extraction and pattern recognition. In recent years, with the rapid development of data acquisition technology, data-driven methods based on deep learning(DL)have shown significant advantages in the field of complex system fault diagnosis due to their high efficiency and economy [9]. Such methods can make full use of massive operational data to automatically mine the nonlinear mapping relationship between fault features and system states, thus significantly improving the accuracy and generalization ability of diagnosis [10,11]. Among deep learning approaches, CNNs have emerged as particularly effective for bearing defect detection and classification tasks [12,13]. For example, Eren [14] proposed a fast-bearing fault detection system based on 1-D CNN. Liang [15] proposed a hardware implementation of a one-dimensional convolutional neural network with depth-separable convolution (DSC).Ozcan et al. [16] Introduced an innovative diagnostic system employing multi-branch deep convolutional networks for processing bearing vibration signals. Since rolling bearing vibration signals are usually interfered with by noise, Zhang et al. [17]proposed to reconstruct the signal from the mode components of the optimized VMD decomposition as an input to the 1-D CNN to obtain a fault diagnosis model. The signal noise can be eliminated, the state characteristics can be enhanced, and the proposed hybrid method has superior fault diagnosis capability for bearing vibration signals。.

Ince et al. [18] proposed a fast and accurate condition monitoring and early fault detection system using a 1-D convolutional neural network, thereby effectively reducing manual intervention [19]. However, compared with 1-D signals, 2-D image data has a significant advantage in fault feature characterization, and the multidimensional information encoding approach enables convolutional neural networks to more fully utilize their hierarchical feature extraction capabilities. Through sliding convolution operations in the local sensory field, CNNs can adaptively capture discriminative local patterns (e.g., textures, edges, etc.) in the image and fuse them layer by layer to form a global feature representation [20,21]. Consequently, to maximize CNN effectiveness, researchers have extensively investigated diagnostic approaches that transform one-dimensional temporal signals into two-dimensional feature representations. Typical implementations involve converting vibration data into grayscale image formats. [22], recurrence plots(RPs) [23], and recursive quantization analysis(RQA) [24,25] to take into account nonsmooth excitation and nonlinear damage evolution. The Recurrence Plot(RP)is a nonlinear signal processing method developed in recent years to analyze nonsmooth and nonlinear signals. It reflects the dynamic characteristics of the system by mapping the time series into a high-dimensional phase space and demonstrating the self-similarity of the signals across time scales in the form of a two-dimensional image. Compared with traditional linear methods, RP can more effectively capture the nonlinear changes of the system during the evolution process and is more sensitive to the dynamic changes caused by structural damage. Existing studies have shown that the method has been successfully applied in the fields of EEG signal feature extraction, audio signal processing, and structural health monitoring [26,27]. Especially in structural damage identification, RQA can effectively quantify the subtle changes in a system's dynamic properties, providing a new technical approach for early damage detection. Pathirage et al. [28,29] input the intrinsic frequency and vibration mode information of a structure into a deep sparse autoencoder framework for identifying the damage conditions. Therefore, this paper expects to achieve automated and efficient recognition of pattern changes in recurrence plots(RPs)with the help of convolutional neural networks(CNNs), which are widely used in the field of image recognition. Specifically, different damage locations and damage degrees will be reflected as different image patterns in the recurrence plot generated by the acceleration signal. Based on this, CNN can be used to extract damage-sensitive features in the recurrence plot and further classify and recognize them, thereby enabling the automated diagnosis of structural damage.

Although the recurrence plot(RP) [30] method performs well in nonlinear feature extraction, there are still some technical bottlenecks. First, RP is less efficient in processing long-time sequences, and the computation volume rises rapidly with the increase of signal length, making it challenging to extract practical features in long-time sequences. In addition, RP is more sensitive to noise, which easily leads to blurring of image features. In practical applications, rolling bearings operate in a complex and variable environment characterized by strong non-stationarity of vibration signals. These weak early fault signals are easily masked by noise and are more susceptible to noise interference [31,32].CNN and LSTM have become a popular research topic because of their ability to efficiently capture the complex mapping relationships between different signals and fault states [33,34]. Despite the recognized effectiveness of recurrence plots (RP) in analyzing nonlinear dynamic signals, current bearing fault diagnostic approaches still exhibit certain shortcomings. In this paper, a bearing fault diagnosis method based on TFRP and PSF-CNN-BiLSTM with good noise immunity is proposed. Firstly, based on the decreasing information entropy criterion to determine the number of wavelet decomposition layers (EDWPD), the Db4 wavelet basis function is used for WPD decomposition to realize the accurate division of the signal in the key frequency bands, the time delay embedding is introduced to construct the phase space, and the recursive map threshold adaptive algorithm based on the Euclidean distance is designed to maintain the stability of the features. Finally, the extracted features are used as inputs to the PSF-CNN-BiLSTM model to train the neural network for classifying bearing faults. The main contributions of this paper are as follows:

1) Aiming at the problem that traditional wavelet packet decomposition relies on empirical settings in the selection of decomposition layers, this paper proposes an adaptive termination criterion based on the change of information entropy.

2) Based on the EDWPD and RP construction methods, a TFRP feature map is proposed, and a non-thresholding mechanism is introduced to enhance feature stability.

3) A two-channel convolutional encoder is designed as a feature extractor, which utilizes a multi-scale sense field and feature fusion

strategy to strengthen the feature abstraction and expression ability and provide more discriminative feature representation for subsequent classification.

4) Combine the eigenmode features extracted by EVD with PSF-CNN-BiLSTM architecture to construct an end-to-end fault diagnosis model. The model achieves satisfactory classification accuracy on both CWRU and Paderborn datasets, which verifies its practicality and superiority.

This paper proceeds with the following organizational framework: Section II describes the EDWPD and TFRP, as well as the method for extracting eigenmode features. Section III establishes a bearing fault diagnosis model based on PSF-CNN and BiLSTM. In Section IV, The diagnostic performance of the proposed method is comprehensively evaluated through ablation studies and comparative experiments using both CWRU and Paderborn bearing datasets. Finally, the conclusions of this work are drawn in Section V.

2. Feature processing methods

2.1. Entropy decreasing based wavelet packet decomposition

As a generalized form of the traditional wavelet transform, wavelet packet decomposition (WPD) realizes the satisfactory resolution of the signal in the whole frequency band through the multiresolution analysis framework. The core innovation is to break through the limitation of the traditional wavelet decomposition that only recursively decomposes the low-frequency components and to synchronize the decomposition of the high-frequency components by constructing a complete binary tree structure, to form a homogeneous and adaptive division in the time–frequency plane, which is more likely to highlight the details of the features that are submerged in the high-frequency noise. The formula of WPD is as follows:

$$\begin{cases} \varphi(x) = \sqrt{2}\sum_{n\in Z} h_n \varphi(2x-n) \\ \psi(x) = \sqrt{2}\sum_{n\in Z} g_n \varphi(2x-n) \end{cases} \tag{1}$$

where $\varphi(x)\,\psi(x)$ are the scale function and wavelet basis function for multiscale analysis, where h_n denotes the high-pass filter coefficients and g_n corresponds to the low-pass filter coefficients in the decomposition framework,$h_n^2 + g_n^2 = 1$.

Wavelet packet decomposition is employed to separate the input signal into its low-frequency approximation coefficients and high-frequency detail coefficients. For a time series signal of length N,$\{x(t)\}, t \in [1, N]$ the decomposition coefficients are

$$\begin{cases} x_{l+1,2k}^{m} = \sum_{n\in Z} h_{n-2m} x_{l,k}^{n} \\ x_{l+1,2k+1}^{m} = \sum_{n\in Z} g_{n-2m} x_{l,k}^{n} \end{cases} \tag{2}$$

where $x_{l,k}^{m}$ denotes the sequence of wavelet packet coefficients in the k th subband of layer l,n denotes the number of wavelet packet coefficients in the k th subband of layer l,$n = N/2^l$.

Aiming at the limitation that the number of layers of traditional wavelet packet decomposition(WPD)depends on the empirical setting, this paper proposes a dynamic termination criterion based on the difference of information entropy. The method realizes the balance between the completeness of signal feature extraction and computational efficiency by quantifying the entropy change of energy distribution between adjacent decomposition layers. Theoretical analysis shows that when the entropy difference approaches zero, the marginal information gain generated by continued decomposition is significantly reduced, and the termination of decomposition at this time can avoid the risk of overfitting. In this paper, the decreasing information entropy criterion is induced to determine the number of wavelet decomposition layers(EDWPD), and the basic principles are as follows:given a bearing data set $X = \{x_1, x_2, \ldots, x_m\}$, where $\boldsymbol{x}_i = (x_{i,1}, \ldots, x_{i,j}, \ldots, x_{i,n})$ is the one-dimensional vibration acceleration data,m is the total number of data, i is the data serial number, and n is the data dimension. For the i th data x_i, after WPD, the decreasing information entropy of the j th layer is defined as follows:

$$A_{i,j} = E_{i,j} - E_{i,j-1} \tag{3}$$

where $E_{i,j}$ is the average value of information entropy of all frequency bands corresponding to wavelet coefficients in layer of the i th data, that is, the original information entropy criterion, which is defined as follows:

$$E_{i,j} = \left(E_{i,j,1} + E_{i,j,2} + \cdots + E_{i,j,k} + \cdots + E_{i,j,2^j}\right)/2^j \tag{4}$$

where $E_{i,j,k}$ is the sub-information entropy value of the k th band of the j th layer of the i th data

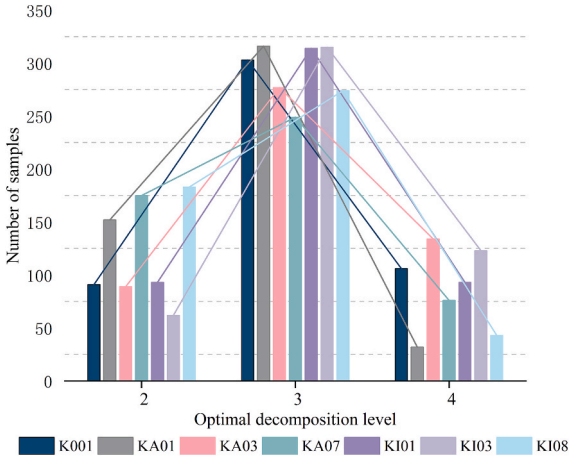


Fig. 1. The distribution of optimal decomposition levels across different fault categories.

$$\begin{cases} E_{j,k} = -\sum_{q=1}^{s} P(W_{i,j,k,q}) \lg\left(P(W_{i,j,k,q})\right) \\ P(w_{i,j,k,q}) = \dfrac{(w_{i,j,k,q})^2}{\sum_i (w_{i,j,k,q})^2} \\ \text{s.t.} \sum_{q=1}^{s} P(w_{i,j,k,q}) = 1 \end{cases} \tag{5}$$

where:$W_{i,j,k,q}$ is the q th wavelet coefficient of the k th band of the j th layer of the i th data, $q = 1, 2, \ldots, s, \quad s = \frac{n}{2^j}$; $P(W_{i,j,k,q})$ is the probability of occurrence of $W_{i,j,k,q}$.On the basis of the decreasing information entropy function described in Eqs. (3) to (5), the final number of decomposition layers of the WPD is determined by the number of layers corresponding to the minimum value of $A_{i,j}$, for the i th data x_i

As shown in Fig. 1, based on the entropy reduction criterion, we automatically terminate decomposition by monitoring the entropy difference between adjacent layers, thereby retaining key frequency band features of the samples while avoiding over-decomposition. In this study, we conducted a statistical analysis on 500 samples for each fault category. First, we calculated the average information entropy of the wavelet coefficients at each layer to measure the information complexity after decomposition. Then, we determined the information saturation point based on the entropy difference between adjacent layers and selected the layer with the smallest entropy difference as the optimal decomposition layer. The results showed that three layers achieved the best balance, and no high-frequency aliasing phenomenon occurred.

2.2. *Fundamentals of Threshold-Free recurrence plot methods*

Recurrence Plot (RP) represents an advanced visualization technique that employs nonlinear dynamics principles to reveal hidden patterns, phase-space relationships, and temporal correlations within univariate time-series data. Takens' theorem is used to analyze the original time series $\{x_1, x_2, \ldots, x_n\}$ Phase space reconstruction at sampling intervals results in

$$X_i = \{x_1, x_2, \ldots, x_n\}, \quad (i = 1, 2, \ldots, N) \quad \xi_2(\xi_2 = 2) \tag{6}$$

where τ represents the delay time and d denotes the embedding dimension, let X_i represent a phase-space vector obtained through time-delay embedding. The distance metric between vectors $\boldsymbol{X}_i$ and $\boldsymbol{Y}_j$ in this reconstructed phase space is computed as:

$$d_{ij} = \left\| \boldsymbol{X}_i - \boldsymbol{Y}_j \right\| \tag{7}$$

ε is a given threshold, Empirical studies suggest setting this threshold below 10 % of the maximum phase-space diameter or alternatively under 25 % of the signal's standard deviation [35], and the recursive value can be expressed as

$$R_{ij}(i, j, \varepsilon) = \Theta(\varepsilon - \| \boldsymbol{X}_i - \boldsymbol{Y}_j \|) \tag{8}$$

$\boldsymbol{X}_i$ is the j th data point of the reconstructed dataset $\{\boldsymbol{X}_e\}$, and $\boldsymbol{Y}_j$ is the j th data point of the reconstructed dataset $\{\boldsymbol{Y}_e\}$, $i,j = 1,2,\ldots,N$, The recurrence plot (RP) is highly sensitive to the selection of time delay, embedding dimension, and threshold. The optimal time delay τ can be estimated using mutual information (MI) [36]

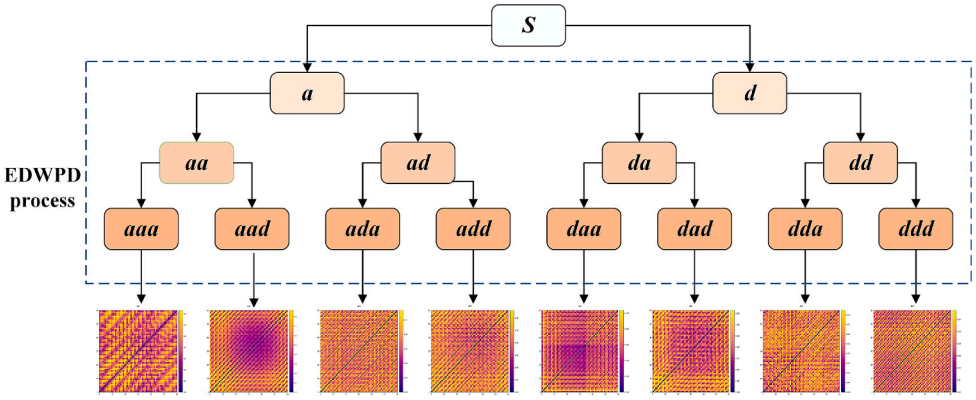


Fig. 2. Process of TFRP based on EDWPD.

$$I = \sum_{ij} p[x_i, x_j(\tau)] \log_2 \frac{p[x_i, x_j(\tau)]}{p[x_i]p[x_j(\tau)]} \tag{9}$$

$p[x_i]$,$p[x_j(\tau)]$, and $p[x_i, x_j(\tau)]$ denote the individual and joint probability densities of time series x_i and $x_j(\tau)$, whereas $x_j(\tau)$ is derived from x_i using a time delay τ, The optimal value of τ is determined by identifying the first local minimum in the mutual information(MI) curve. The embedding dimension d can be chosen by pseudo-nearest neighbor (FNN) [37]. This method examines how nearest neighbors evolve as the embedding dimension shifts from d to $d+1$, with neighbor selection governed by two criteria.

$$\frac{\sqrt{\left\|X_i - X_i^{(r)}\right\|_{d+1}^2 - \left\|X_i - X_i^{(r)}\right\|_d^2}}{\left\|X_i - X_i^{(r)}\right\|_d} > \xi_1 \quad \text{and} \quad \frac{\left\|X_i - X_i^{(r)}\right\|_{d+1}}{\sigma_x} > \xi_2 \tag{10}$$

$\left\|X_i - X_i^{(r)}\right\|_d$ represents the Euclidean distance separating phase point X_i from its nearest neighbor $X_i^{(r)}$ within a $d-$dimensional space, while σ_x corresponds to the standard deviation of x.$\xi_1(10 \leqslant \xi_1 \leqslant 50)$ and $\xi_2(\xi_2 = 2)$ are thresholds. The optimal value of d is reached when nearest neighbors diminish to zero with increasing embedding dimensions.$\Theta(\cdot)$ is the Heaviside function with the expression

$$\Theta(\cdot) = \begin{cases} 1, & x \geqslant 0 \\ 0, & x < 0 \end{cases} \tag{11}$$

R_{ij} represents the recurrence value (0for white points,1for black points), while R_{ij} is visualized on a coordinate plot with i and j as the horizontal and vertical axes respectively. where i and j are the time series labels respectively, and the resulting image is the recurrence map. The traditional recurrence map with thresholding is prone to the loss of some detailed features due to the introduction of fixed thresholding in the construction process. The Threshold-Free Recurrence Plot (TFRP), on the other hand, abandons the binarization of 0/1, and is able to present smoother gradient changes, reflecting the nonlinear dynamics of the system. TFRP has a stronger abstract expression ability in preserving the deeper characteristics of the original signal attributes. Therefore, in this paper, Threshold-Free Recurrence Plot is chosen as the feature representation to enhance the distinguishability and expression ability of the structural damage patterns, and for the data $\boldsymbol{X}_i$ and $\boldsymbol{Y}_j$, the matrix elements in the TFRP are defined as follows:

$$R(i,j) = \|\boldsymbol{X}_i - \boldsymbol{Y}_j\| \tag{12}$$

In rolling bearing vibration analysis, excessively long signal durations can lead to blurred TFRP features, with noise obscuring the subtle details of the original signal. To address this, this study determined the number of wavelet decomposition layers (EDWPD) to be three by reducing the information entropy criterion, and employed the Db4 wavelet for three-layer decomposition, converting the subband coefficients into TFRP. The Db4 wavelet effectively captures transient impacts and high-frequency resonance characteristics in bearing fault signals, reduces phase distortion, and outperforms asymmetric wavelets (such as Symlets). Compared to lower-order Db wavelets (e.g., Db2), which lack sensitivity to weak features, and higher-order Db wavelets (Db8–Db10), which significantly increase computational costs, Db4 achieves the optimal balance between feature extraction capability and computational efficiency.

Fig. 2 shows the construction process of TFRP, where S is the original signal, 'a' denotes the low-frequency approximation (approximation), and 'd' denotes the high-frequency details, e.g., 'aaa': 3 consecutive low-frequency decompositions (the lowest frequency sub-band) 'aad': the first two low-frequency + the last high-frequency decompositions, and 'ddd':3 consecutive high-frequency decompositions (the highest frequency sub-band) as shown in Fig. 1, as the number of WPD layers increases, the generated sequence of approximation coefficients and detail coefficients becomes shorter, which solves the problem that TFRP is difficult to reflect the long-term sequence characteristics. The spectral disparity between noise components and fault signatures enables wavelet packet decomposition (WPD) to effectively segregate fault-related features from background noise. Through its multi-resolution analysis framework, WPD generates decomposition coefficients that preserve essential signal characteristics across multiple scales.

Table 1
EDWPD and TFRP generation time for each sample.

Steps	EDWPD	TFRP
Time(s)	0.04	0.237

Consequently, the derived TFRP achieves dual benefits: (1) comprehensive representation of signal features at varying resolutions, and (2) inherent robustness against noise interference.

Table 1 shows the time required to generate EDWPD and TFRP for each sample.

2.3. TFRP-EVD structure

TFRP is a real symmetric matrix, so it can definitely be decomposed into eigenvalues. Let $U \in R^{n\times n}$ be a $n-$order real symmetric matrix. The eigenvalue decomposition of matrix A is to find an orthogonal matrix $U \in R^{n\times n}$ such that

$$A = U\Lambda U^T \tag{13}$$

Where U^T is the transpose of U, and Λ is a diagonal matrix.$\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, $\lambda_i, i = 1, \ldots, n$.

TFRP comprehensively characterizes the dynamic evolution process of a system by constructing a complete distance matrix. Its spectral analysis based on EVD can effectively extract the nonlinear dynamic characteristics of the system: the dominant eigenvalues reflect the energy distribution of the core motion modes of the system, the intermediate eigenvalues reveal transient coupling effects and nonlinear harmonic components, and the small eigenvalues characterize dynamic noise. Sudden changes, decay, or clustering of eigenvalues can respectively indicate dynamic phase transitions or quasi-periodic behavior. There are two reasons for choosing the TFRP feature decomposition: 1. Eliminate dependence on threshold selection. Traditional recursive graphs (RP) require manual threshold setting (such as fixed values or dynamic thresholds), and threshold selection significantly affects the robustness of feature extraction. Too small a threshold causes noise to be misjudged as valid signals. Too large a threshold causes weak dynamic features to be ignored. The improvement of TFRP is based directly on the distance matrix of the signal (without binarization), and global dynamic features are extracted through eigenvalue decomposition (EVD), completely avoiding the threshold selection problem. 2. Preserving the complete dynamic information of the original system: Traditional RP loses amplitude information in the distance matrix through binarization (0/1 matrix), while TFRP retains the complete dynamic characteristics. Larger eigenvalues typically correspond to the dominant dynamic components of the signal, while smaller eigenvalues are often associated with background noise or secondary dynamic processes.

3. Proposed bearing fault diagnosis model with Parallel-Scale-Fusion CNN and BiLSTM

Leveraging CNNs' established capabilities in visual pattern recognition, this study develops a novel bearing fault diagnosis framework integrating Parallel-Scale-Fusion convolutional networks with BiLSTM architecture.

3.1. Proposed Parallel-Scale-Fusion CNN

Characterized by localized receptive fields, shared weights, and progressive dimensionality reduction, CNNs possess unique advantages in visual data processing. Unlike traditional approaches requiring explicit feature extraction, CNNs automatically learn hierarchical representations through their layered structure of convolution, nonlinear activation, pooling, and classification components.

1) Input Layer: the input layer is the starting point of the whole model, which is responsible for receiving raw data and passing it to the subsequent convolutional layer for feature extraction.

2) The convolutional layer employs a local receptive field mechanism to perform hierarchical feature extraction on input sequences, enabling the network to capture intricate patterns across different abstraction levels. In the context of one-dimensional feature sequences—such as raw sensor signals or statistical feature vectors—shallow convolutional operations typically detect low-level patterns, including abrupt transitions (e.g., edges in signals), periodic fluctuations, or transient anomalies. In contrast, deeper convolutional layers progressively integrate these primitive features into higher-level semantic representations, facilitating discriminative characterization of distinct failure modes or signal classes. For each feature map, the convolution computes the expression as follows:

$$x_j^l = f\left(\sum_{i\in M_j} x_i^{l-1} * \omega_{ij}^l + b_j^l\right) \tag{14}$$

l represents the l th network layer, while i and j denote the feature map indices for layers l and $(l-1)$ respectively;* denotes the convolution operation,M_j represents the feature map from the preceding layer.ω_{ij}^l specifies the convolution kernel parameters that transform the j th feature map of layer l into the $l-1$ th feature map of layer i;b^l represents the bias term associated with x_i^l, which

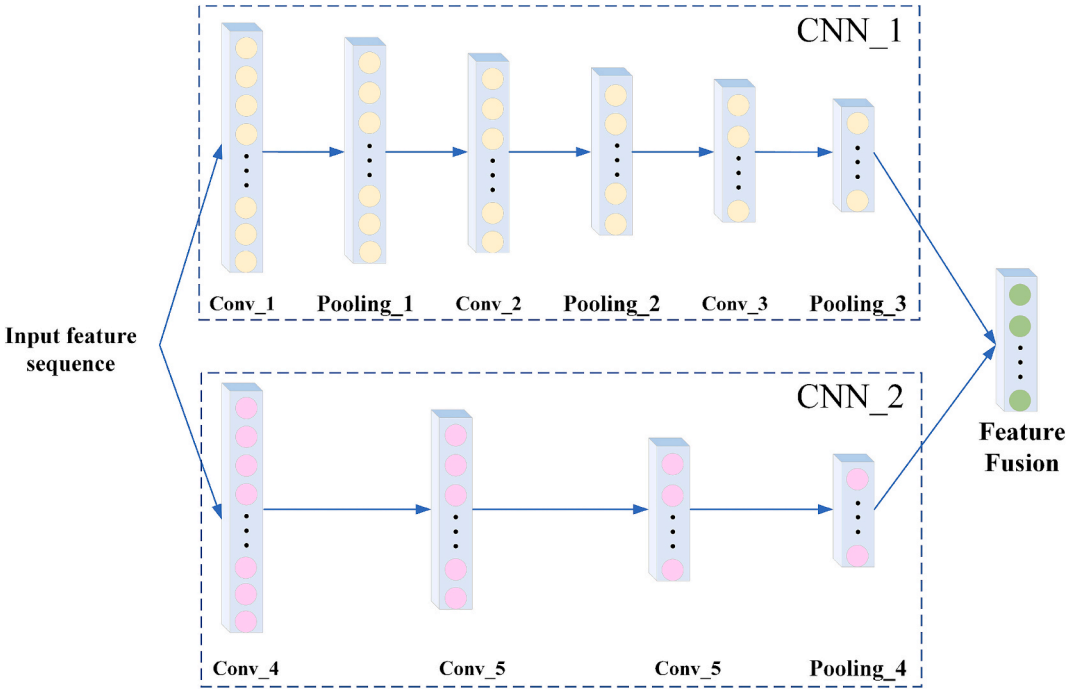


Fig. 3. Parallel-Scale-Fusion CNN model.

denotes the l th feature map in the j th network layer. In the feature map convolution process, a convolution kernel of $f \times f$ is used to convolve the feature map of size $h \times w$. The convolution stride is fixed at S, yielding an output feature map dimension of:

$$h' = \lfloor \frac{h - f + S}{S} \rfloor, \quad w' = \lfloor \frac{w - f + S}{S} \rfloor \tag{15}$$

where $\lfloor \cdot \rfloor$ represents the rounding function, while h' and w' correspond to the post-convolution feature map dimensions (height and width respectively).

3) The pooling layer constitutes an essential CNN component designed to downsample feature maps produced by convolutional operations; this operation reduces computational complexity while preserving the translation-invariant characteristics of learned features. The computation of the pooled feature map is:

$$x_j^l = f\left(\beta_j^l \cdot \mathrm{down}\left(x_j^{l-1}\right) + b_i^l\right) \tag{16}$$

$\mathrm{down}(\cdot)$ denotes the downsampling function for feature map size reduction, while β_j^l and b_i^l represent the first layer's weight matrix and bias vector, respectively."

4) The convolutional and pooling layers serve as feature extraction modules within the network, capturing hierarchical and spatially invariant representations from input images. In contrast, the fully connected (FC) layer functions as a feature classification component, mapping the extracted high-level features to class-specific predictions. The expression of fully connected layer is

$$a^l = \sigma\left(W^l \cdot a^{l-1} + b^l\right) \tag{17}$$

a^l represents the nonlinear output of the fully-connected layer, derived from previous layer output a^{l-1} through weighted transformation (weights W^l, bias b^l) and activation function $\sigma(\cdot)$.

5) The output layer receives activations from the final fully-connected layer, typically transforming them through a Softmax classifier for prediction generation.".

In this study, in order to extract multi-level and multi-scale feature information from raw signals, a two-channel multi-scale convolutional encoder structure is designed to expand the sensory field layer by layer and enhance the feature abstraction capability. Each channel uses a different size of the convolutional kernel and step size: CNN_1 obtains local fine-grained features with a smaller convolutional kernel and shallow pooling operation, and the other channel, CNN_2, realizes the extraction of global semantic information with a larger convolutional kernel and step size. The outputs of the two channels are fused using a product fusion strategy to achieve feature coupling, as this better simulates the physical modulation relationships in bearing faults, thereby obtaining a joint multi-scale feature representation and forming a multi-layer convolutional neural network structure. The Parallel-Scale-Fusion CNN

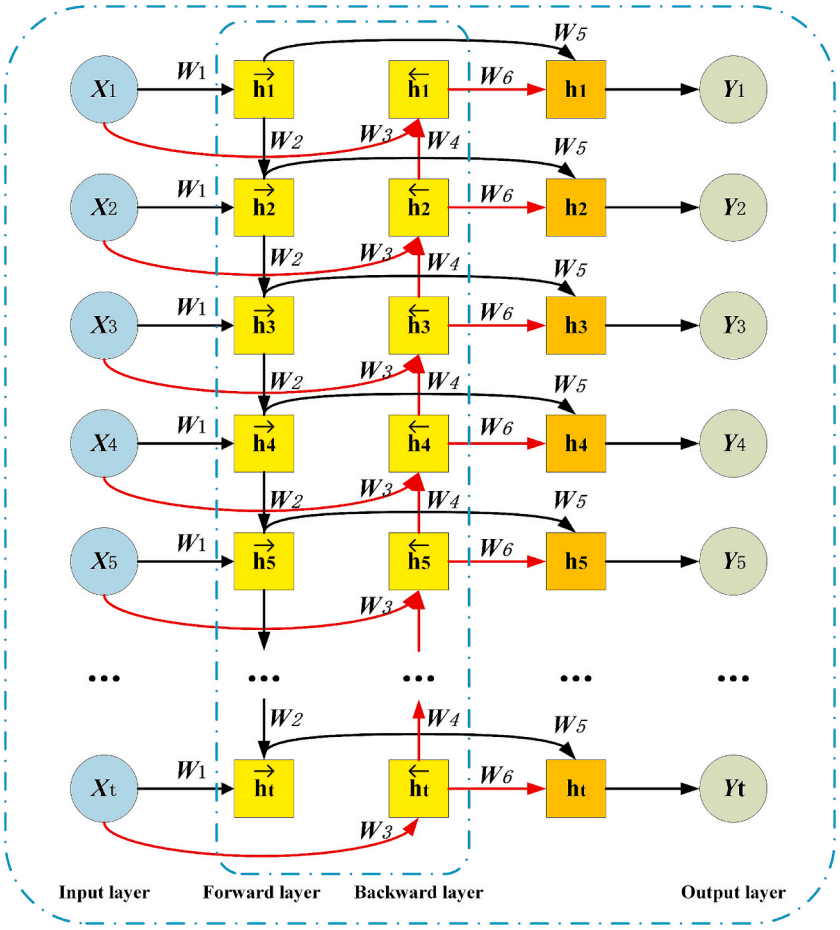


Fig. 4. Structure of BiLSTM network.

model proposed in this study is shown in Fig. 3.

3.2. Introduction to Bidirectional long and Short-Term memory networks

While LSTMs overcome traditional RNNs' limitations in capturing long-term temporal relationships, their sequential processing nature restricts them to historical context only. BiLSTM networks resolve this constraint through parallel forward and backward processing streams, allowing simultaneous consideration of past and future sequence information. This dual-directional analysis provides a more robust feature representation, which is particularly beneficial for time-series prediction and analysis applications.

Fig. 4 illustrates the BiLSTM architecture, where temporal processing occurs bidirectionally through complementary computational pathways. The forward layer sequentially processes input data while maintaining hidden state representations, while the backward layer executes inverse temporal processing. These dual pathways ultimately merge through a concatenation operation, producing comprehensive context-aware outputs according to the following formulation:

$$\begin{cases} \overrightarrow{h}_t = f\left(w_1 x_t + w_2 \overrightarrow{h}_{t-1}\right), \\ \overleftarrow{h}_t = f\left(w_3 x_t + w_4 \overleftarrow{h}_{t+1}\right), \\ y_t = g\left(w_5 \overrightarrow{h}_t + w_6 \overleftarrow{h}_t\right), \end{cases} \tag{18}$$

$\overrightarrow{h}_t$ and $\overleftarrow{h}_t$ represent the respective outputs of the forward and inverse LSTM pathways, while y_t denotes the hidden layer's final output, obtained through activation functions $f(\cdot)$ and $g(\cdot)$.

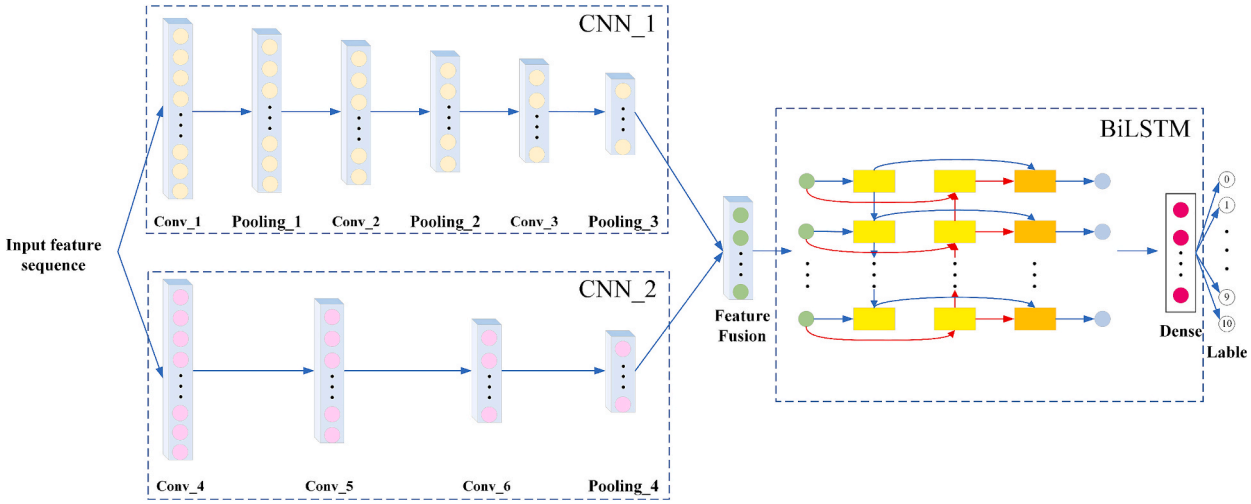


Fig. 5. Structure of the proposed Parallel-Scale-Fusion CNN and BiLSTM model.

Table 2
Configuration of the proposed Parallel-Scale-Fusion CNN and BiLSTM model.

Structure	No.	Layers	Description of key parameters
PSF-CNN	1	Conv1D_1	filters = 50,kernel_size = 5,strides = 1
	2	MaxPooling1D_1	pool_size = 2 strides = 2
	3	Conv1D_2	filters = 40,kernel_size = 5,strides = 1
	4	MaxPooling1D_2	pool_size = 2, strides = 2
	5	Conv1D_3	filters = 30,kernel_size = 4,strides = 1
	6	MaxPooling1D_3	pool_size = 3, strides = 4
	7	Conv1D_4	filters = 50,kernel_size = 7,strides = 2
	8	Conv1D_5	filters = 40,kernel_size = 6,strides = 1
	9	Conv1D_6	filters = 30,kernel_size = 3,strides = 2
	10	MaxPooling1D_4	pool_size = 4, strides = 4
	11	Feature Fusion	
BiLSTM	12	Bidirectional-LSTM_1	units = 128
	13	Bidirectional-LSTM_2	units = 128
Fully Connected Layer	14	Dropout	rate = 0.5
	15	Dense	Output fault category probability

3.3. Parallel-Scale-Fusion CNN and BiLSTM models

In order to fully exploit the multilevel temporal features at both local and global scales, as shown in Fig. 5, this paper constructs a two-path parallel convolutional feature extraction structure. Table 2 shows the configuration of the proposed PSF-CNN-BiLSTM network. The module contains two parallel encoding paths focusing on fine-grained and coarse-grained feature modeling, respectively, and realizes the coexpression of multiscale information through the tensor-level fusion strategy. Path1: Local-Scale Feature Extraction This path uses a multilayer convolution-pooling structure, which contains three sets of convolution and downsampling operations. The convolution kernel size is small, aiming to maintain a high temporal resolution. Through layer-by-layer stacking, the time dimension is compressed step by step, and discriminative local timing features are extracted to effectively characterize small fluctuations and local structural changes in the signal. Path2: Global-Scale Feature Extraction, which focuses more on understanding the macroscopic structure of the signal. By introducing a large convolutional kernel and stepwise convolution, we realize fast downsampling of the original sequence and construct a coarse-grained global feature representation. The structure can effectively capture longer-time dependencies, which helps to recognize long-period, low-frequency failure modes.

In order to fuse the information of both local and global scales, this paper adopts the Tensor-Level Fusion (TLF) strategy to integrate the two paths. This operation realizes the interactive enhancement of features at different scales, which strengthens the feature expression capability while maintaining the original spatial structure and provides richer and more discriminative feature representations for the subsequent classifiers. The fused multi-scale feature tensor is further fed into a bidirectional long and short-term memory network for mining the contextual dependencies present in the sequences.

Layer 1 BiLSTM: Deep temporal modeling setup outputs the hidden states of all time steps to preserve the complete temporal dynamic information and enhance the sequence context modeling capability. Layer 2 BiLSTM: Timing Compression and Feature Aggregation In order to extract the compressed global timing features, i.e., only the hidden state vectors of the last moments are outputted, this process achieves feature aggregation of the overall time series and extracts the global semantic representations for

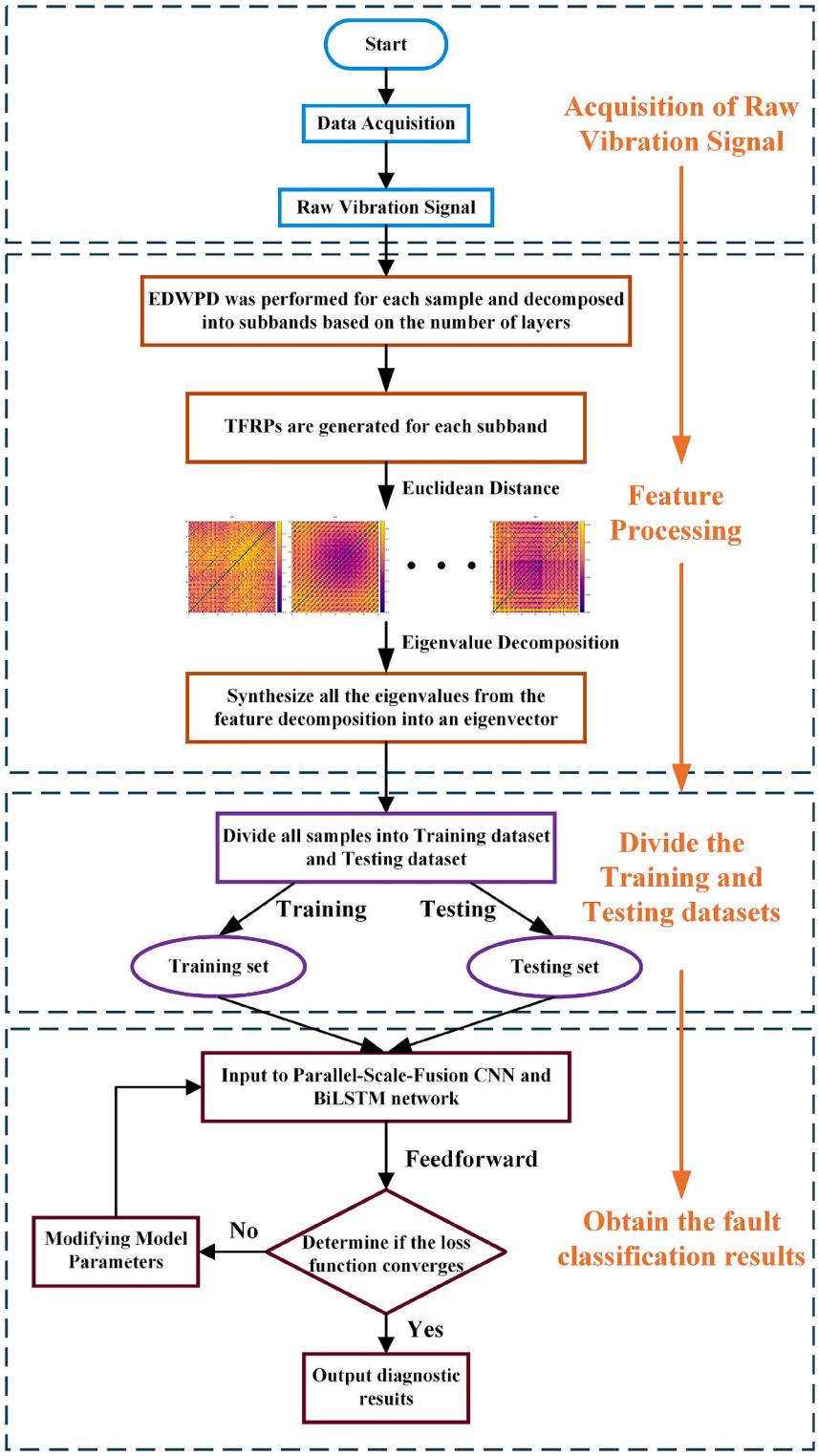


Fig. 6. Flowchart of the proposed method.

subsequent classification. To alleviate model overfitting, this paper adds a Dropout layer for random deactivation after BiLSTM output, with probability set to 0.5. Finally, the network outputs the fault categories through the fully connected layer and Softmax classifier。 The CNN encoder actively filters the correlations between features after EVD through large-span pooling. The current time step design

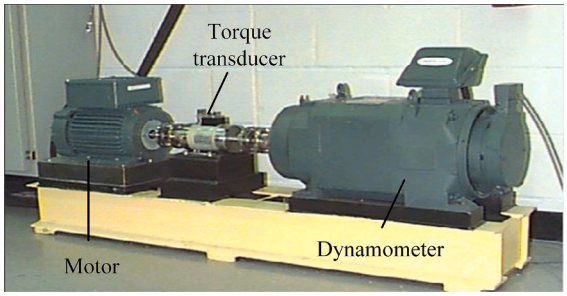


Fig. 7. Test rig of CWRU [43].

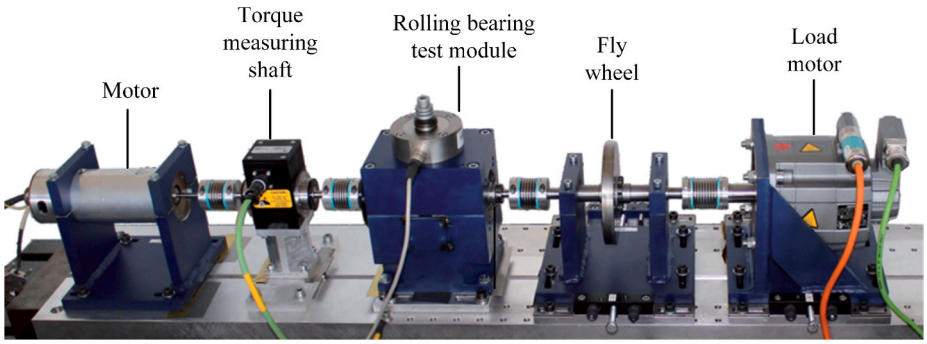


Fig. 8. Test rig of Paderborn [42].

retains fault-sensitive cross-band features and improves computational efficiency.

3.4. Fault diagnosis model proposed in this study

As shown in Fig. 6, this paper aims to fully extract the dynamic characteristics and nonlinear features of the signal, and the method flow is as follows: firstly, the optimal number of layers of wavelet packet decomposition is adaptively determined according to the decreasing information entropy criterion (EDWPD), and the original vibration signal is decomposed into multiple sub-band signals, so as to realize the expression of information at different frequency scales. Subsequently, time-delay embedding is performed for each subband signal to reconstruct its phase space trajectory matrix, and then the corresponding threshold-free recurrence plot (TFRP) matrix is constructed by calculating the Euclidean distances between the trajectory vectors, which describes the nonlinear evolutionary behaviors of the signal in the phase space.

Based on this, features are extracted, and an eigenvalue decomposition (EVD) is performed on the TFRP matrix to extract its eigenvalues as discriminative features, characterizing the signal as an intrinsic dynamic mode, thus constituting the final eigenvector. This feature vector is then inputted into the PSF-CNN-BiLSTM model constructed in this paper for training and testing validation. The experimental results show that the method not only has good nonlinear modeling capability but also realizes effective compression and representation of high-dimensional features while maintaining effective discriminative performance, showing strong robustness and classification performance.

4. Experimental results

In order to validate the effectiveness of the model proposed in this study, we conducted ablation experiments based on the CWRU and Paderborn datasets, and then compared them with previous state-of-the-art methods. The experiments were run on i9-14900HX CPU, RTX 4070 Laptop GPU, 32G RAM, and the nonlinear activation functions tanh and Relu were applied to PSF-CNN convolutional layer and BiLSTM, respectively. We used dropout to avoid overfitting framework after BiLSTM during training, and Adam's algorithm was used as the optimization algorithm that 50 epochs are trained.

Fig. 7 shows the experimental rig used to generate rolling bearing failure data. The experimental rig is dominated by a 2 hp (1.5 KW) motor, a torque transducer, and a power test meter. Deep groove ball bearings (6205-2RS JEM SKF) are mounted on the shaft of the 1.5KW motor, along with various electronic control devices not specifically shown in the figure.

The modular test bench is shown in Fig. 8.The experimental bench adopts a modular structure to facilitate the flexible configuration of different mechanical failure simulation scenarios and is suitable for the generation of failure data for electromechanical drive systems, and consists of five modules: 1) motor; 2) torque measurement axis; 3) rolling bearing test module; 4) flywheel; and 5) load motor. The test object is a type 6203 ball bearing.

Table 3
Sample dataset of artificially damaged bearings.

Bearing Status	Label	Bearing Code	Damage Size	Samples
Normal	0	Normal	None	300
B	1	B007	0.007	300
B	2	B014	0.014	300
B	3	B021	0.021	300
IR	4	IR007	0.007	300
IR	5	IR014	0.014	300
IR	6	IR021	0.021	300
OR	7	OR007	0.007	300
OR	8	OR014	0.014	300
OR	9	OR021	0.021	300

B:Rolling element damage IR:Inner ring damage OR:Outer ring damage Damage size in inch.

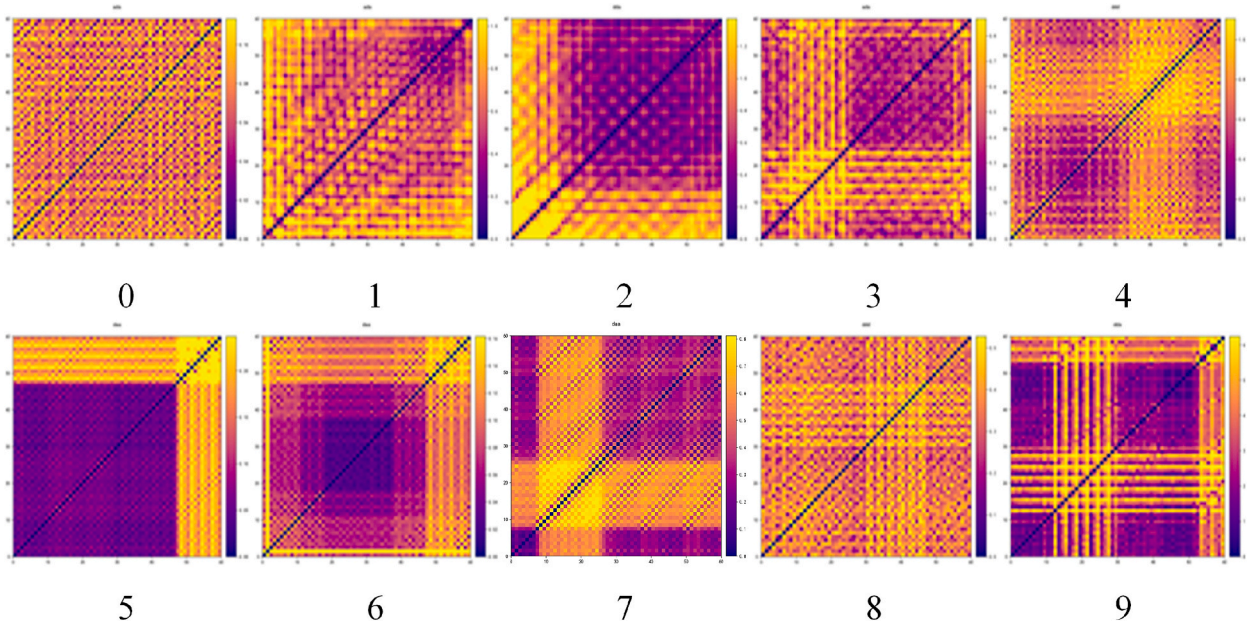


Fig. 9. TFRP feature maps for normal and fault generation for the CWRU dataset.

4.1. CWRU bearing data experimental setup

Effective neural network training and evaluation typically demand substantial datasets, yet practical bearing fault diagnosis scenarios frequently suffer from insufficient sample availability. In order to solve this problem, this paper uses CWRU bearing drive-side data, sampling frequency 12KHz. Motor load 1 horsepower (HP imperial) 0.7457 kW (KW), the motor speed 1797r/min, the motor rotates a circle of about 400 data points, I here to take 400 points as a sample, can completely show the fault cycle, too much data affects the speed of training, so as to balance the model convergence speed and feature learning ability. The proposed approach simultaneously optimizes both convergence behavior and representational learning capacity during model training. The classification includes normal, and nine types of faults, and the faults include rolling body faults, inner ring faults, and outer ring faults. In each fault state, 300 samples were collected by non-overlapping sampling.

Table 3 presents the fault sample set specifications. To ensure model convergence, the training cycle number was fixed at 50 iterations. Comparative experiments evaluated two approaches: (1) ablation studies, and (2) the proposed model, demonstrating its performance advantages.

TFRP can clearly present the timing patterns and structural characteristics of signals in two-dimensional space, with good visualization ability, which helps to identify the dynamic differences between different working conditions. The RP, without setting a threshold, avoids the loss of information caused by artificial parameter selection, can completely retain the trajectory evolution characteristics of the timing signal in the phase space, and shows the obvious image texture differences in different fault states, with strong inter-class separability and discriminative properties, Fig. 9 displays TFRP feature maps generated from normal and nine fault types.

Fig. 10 shows the confusion matrix of 10 trajectories for the conventional TFRP model and ablation experiments and also for the

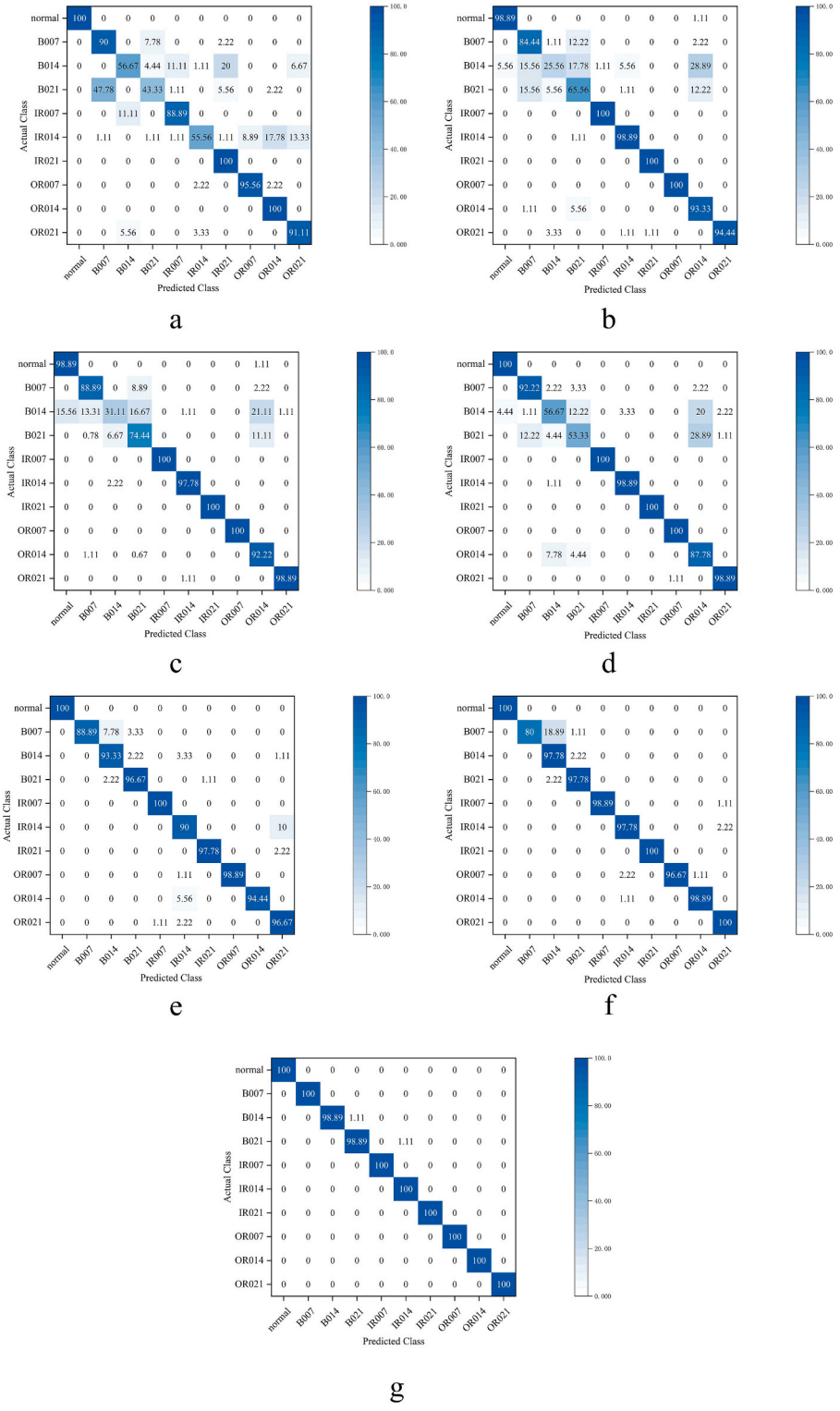


Fig. 10. Confusion matrix based on the TFRP model and the model proposed in this study(a)signal-CNN-BiLSTM;(b)TFRP-EVD-CNN;(c)EDWPD-TFRP-EVD-SVM;(d)TFRP-EVD-CNN-BiLSTM;(e)EDWPD-TFRP-CNN;(f)EDWPD-TFRP-CNN-BiLSTM;(g)The proposed model.

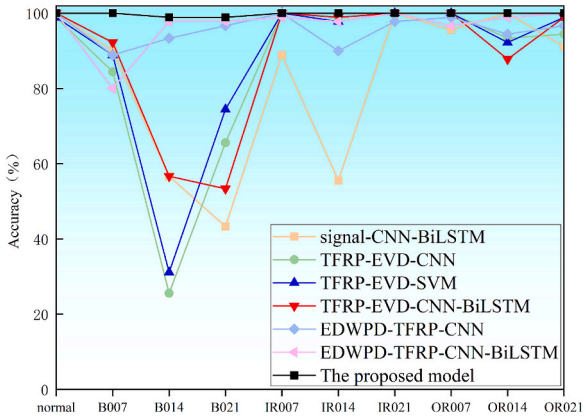


Fig. 11. Accuracy of category predictions on different models.

Table 4
Test training accuracy based on the TFRP model and the model proposed in this study.

Model	Training Accuracy	Testing Accuracy	Dropout Rate	Epoch	Batch size	Time(s) per epoch
signal-CNN-BiLSTM	81.27 %	82.11 %	0.5	50	64	3.4
TFRP-EVD-CNN	89.89 %	86.11 %	0.5	50	64	3.59
TFRP-EVD-CNN-BiLSTM	91.14 %	88.78 %	0.5	50	64	3.87
EDWPD-TFRP-CNN	94.33 %	95.77 %	0.5	50	64	11.98
EDWPD-TFRP-CNN-BiLSTM	97.24 %	96.78 %	0.5	50	64	12.31
The proposed model	100 %	99.78 %	0.5	50	64	4.39

Table 5
Precision rate Recall rate and F1-score for 10 classifications.

Class	Precision(%)	Recall(%)	F1-score(%)
Normal	100	100	100
B007	100	100	100
B014	100	98.89	99.44
B021	98.89	98.89	98.89
IR007	100	100	100
IR014	98.9	100	99.45
IR021	100	100	100
OR007	100	100	100
OR014	100	100	100
OR021	100	100	100
Global Average	99.78	99.78	99.78

proposed model. The highest accuracy of the confusion matrix for the classification of the proposed model approach is 100 %, and the lowest accuracy is 98.89 %, showing that the accuracy of fault classification of the proposed TFRP fault diagnosis model is higher than that of all other conventional models. Fig. 11 shows the accuracy of category prediction on different models, and Table 4 shows the comparison between the proposed model and other models. The results show that the training accuracy is 100 %, and the testing accuracy is above 99 %. Noise introduces large fluctuations in the signal embedding space, resulting in an unstable distance matrix, whereas in this paper, EDWPD is used to remove the noise from the original vibration signals to improve the noise immunity of the proposed fault diagnostic model, and then the features are extracted to perform the training in a way that does not produce this phenomenon.

Computational complexity is quantified via Floating Point Operations (FLOPs), measuring the total arithmetic operations required for complete forward and backward propagation of input images. The Adam optimizer was employed for model training. Performance evaluation incorporated both macro- and micro-averaged metrics derived from the confusion matrix, including precision, recall, F1-score, and overall classification accuracy.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{19}$$

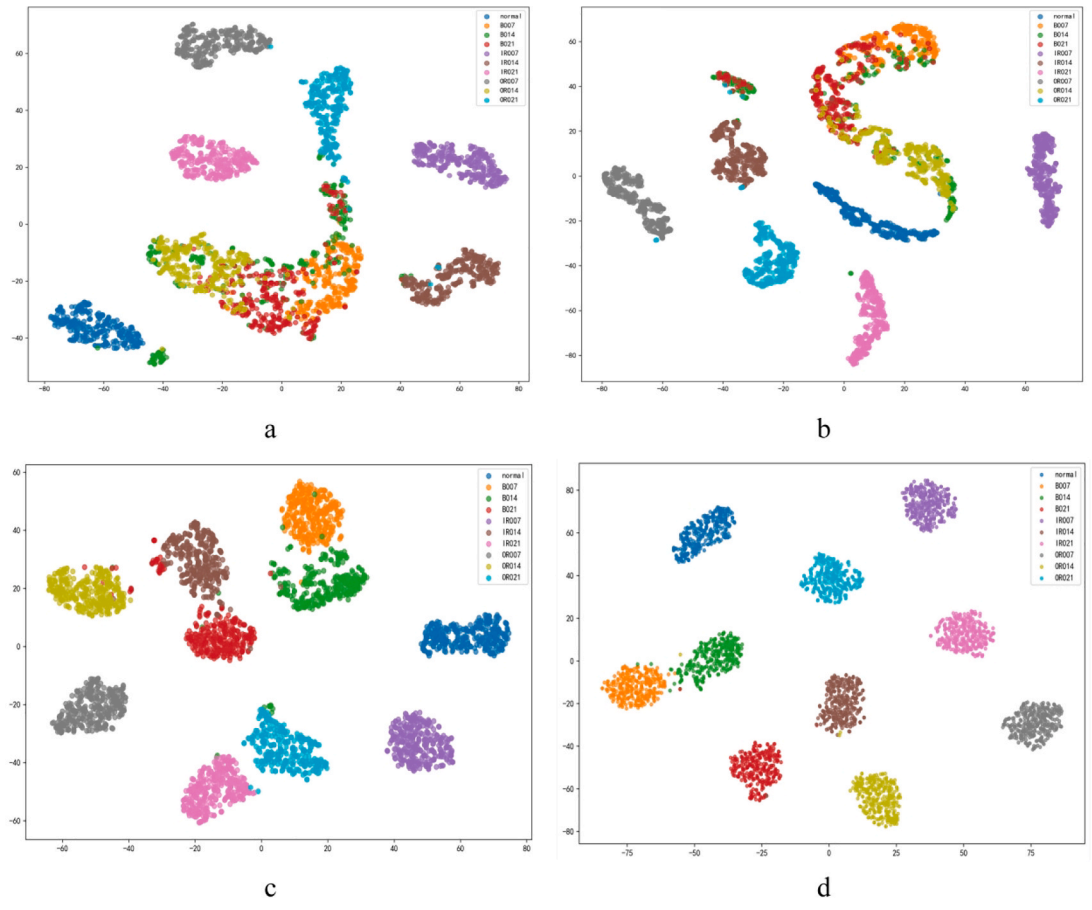


Fig. 12. Deep feature distribution of the feature map based on the TFRP model and the model proposed in this study(a) TFRP-EVD-CNN;(b) TFRP-EVD-CNN-BiLSTM;(c) EDWPD-TFRP-CNN-BiLSTM;(d)The proposed model.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{20}$$

$$\text{F1 - score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{21}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{22}$$

The formulas are shown in Eqs. (19)–(22), and their results are shown in Table 5. The average values of precision, recall, and macro F1 score reach 99.78 %, which indicates that the fusion of PSF-CNN and BiLSTM helps to improve the accuracy, and the present model can achieve high precision. Comparative analysis reveals the PSF-CNN-BiLSTM framework's consistent superiority over baseline methods, demonstrating significantly higher classification accuracy under all experimental conditions.

In order to visualize and further compare the advantages of the proposed method in feature classification, the t-SNE algorithm is used to visualize the high-dimensional deep abstract features extracted from the training model by mapping them to a 2D plane. The test samples are downscaled and displayed as scattered points in a 2D plot, as shown in Fig. 12. In this plot, the horizontal and vertical coordinates represent the normalized values of the output layer of the network, and the colors are used to distinguish the fault types.

4.2. Paderborn bearing data experimental setup

Data set N15_M01_F10 with normal signals and artificial faults (speed of 1500r/min, torque of 0.1Nm, and radial force of 1000 N) are used to validate the effectiveness of the proposed model. The man-made faults include different treatments for inner-ring faults and outer-ring faults. The damage degree level is categorized into level 1 and level 2 with a sampling frequency of 64KHz; when the bearing rotates one revolution at 1500r/min, about 2560 sampling points are collected, so 2560 data points are taken as samples. In each fault condition, 500 samples were collected by non-overlapping sampling.

Fig. 13 displays TFRP feature maps generated by normal and six fault types. Table 6 shows the details of the faulty sample set using

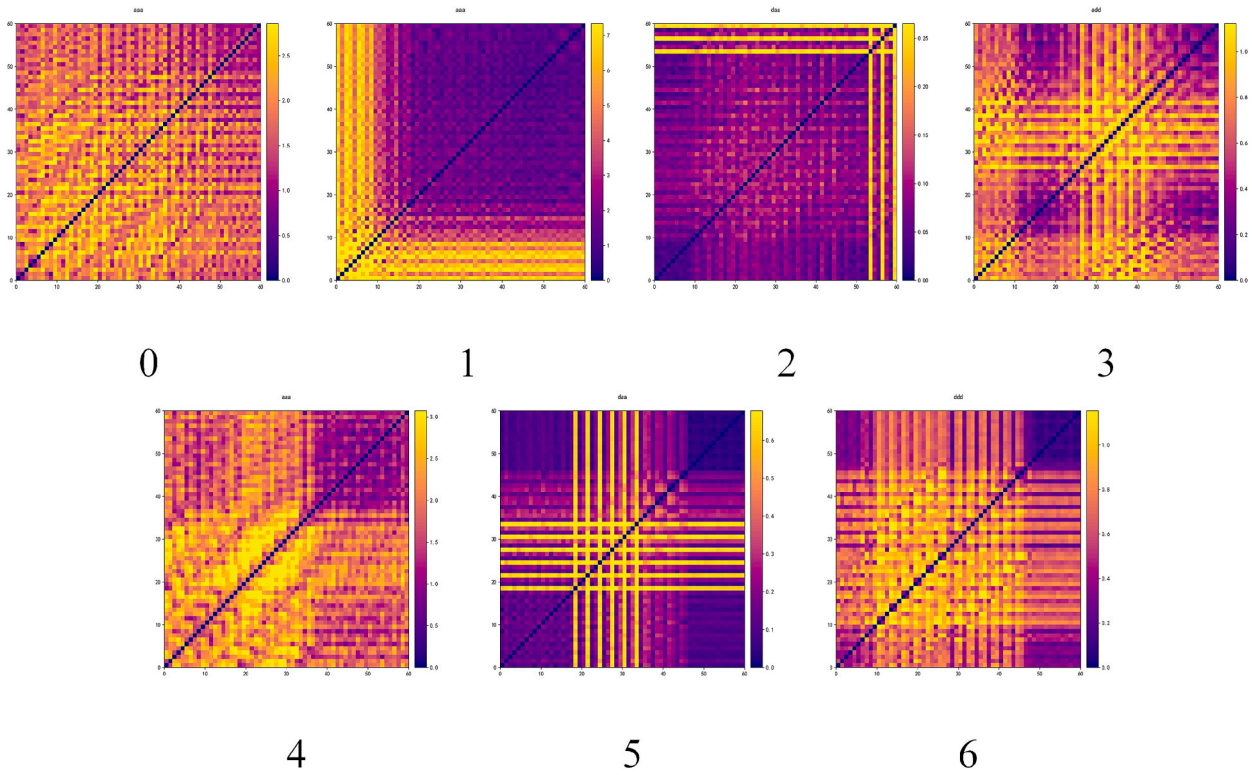


Fig. 13. TFRP Feature Maps for Normal and Fault Generation for the Paderborn Dataset.

Table 6
Paderborn sample dataset of artificially damaged bearings.

Bearing Status	Label	Bearing Code	Damage Method	Extent of Damage	Samples
Normal	0	K001	None	None	500
OR	1	KA01	Electric discharge machining	1	500
OR	2	KA03	Electric engraver	2	500
OR	3	KA07	Drilling	1	500
IR	4	KI01	Electric discharge machining	1	500
IR	5	KI03	Electric engraver	1	500
IR	6	KI08	Electric engraver	2	500

OR:Outer ring damage IR:Inner ring damage Extent of Damage(level)

different damage methods like Electric discharge machining, Electric engraver, Drilling, and different damage levels with the number of training cycles set to 50 rounds.

Fig. 14 shows the confusion matrix and plotted depth feature distribution for the seven fault classifications in the Paderborn dataset. The highest accuracy of the confusion matrix classified by the proposed modeling method is 100 %, and the lowest accuracy is 99.33 %, showing that the fault classification of the proposed TFRP fault diagnosis model fault classification is effective. All the samples except drilled holes are recognized with 100 %. Table 7 shows that the mean values of precision, recall, and macro F1 score reached 99.9 %, respectively. Therefore, the method in this study has the highest ability to recognize different rolling bearing faults and the strongest resistance to noise, which is attributed to the fact that the noise in the signal has been removed by EDWPD.

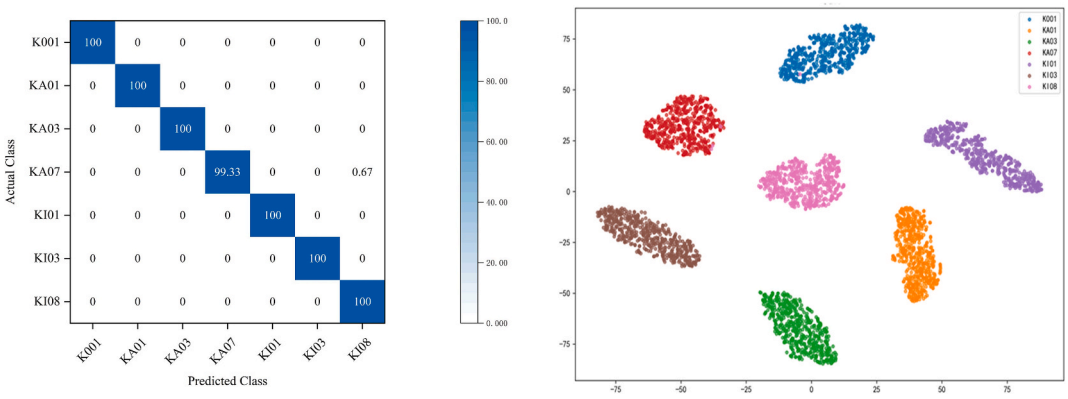


Fig. 14. Confusion matrix and plotted depth feature distribution for 7 fault classifications.

Table 7
Precision rate Recall rate and F1-score for 10 classifications.

Class	Precision(%)	Recall(%)	F1-score(%)
K001	100	100	100
KA01	100	100	100
KA03	100	100	100
KA07	100	99.3	99.66
KI01	100	100	100
KI03	100	100	100
KI08	99.34	100	99.67
Global Average	99.9	99.9	99.9

Table 8
Accuracy of existing methods.

Method	Accuracy
VSI-DGCAPN	98.19 %
GAF-CNN	99.82 %
NAS-PERIRB	99.43 %
IFM-based ResNet	99.70 %
AMDC-CNN	99.80 %
The proposed model	99.90 %

4.3. Comparison of existing models

The Paderborn dataset creators developed a comprehensive diagnostic approach combining feature engineering (extraction and selection) with ensemble classification techniques for precise fault categorization. This section conducts comparative analysis between the proposed model and state-of-the-art methods with optimal classification accuracy, validating the model's effectiveness and competitive advantages. J. Cao [38] et al. developed a neural architecture search (NAS) framework incorporating two key innovations: (1) an early stopping mechanism and (2) an Insert Replay Buffer (IRB) for computational efficiency enhancement. The method, termed NAS-PERIRB, further integrates a Pareto-optimal reward function to simultaneously optimize model accuracy. L. Hou [39] et al. proposed an Input Feature Map (IFM) method combined with residual networks, which is capable of automatically extracting effective features without the need of preset parameters, thus improve the diagnostic performance. D. Wang [40] et al., on the other hand, designed a multidimensional cascaded convolutional neural network (AMDC-CNN) based on the attention mechanism, which strengthens the important features through the attention mechanism while fusing the multidimensional vibration and torque signals to achieve a higher fault recognition accuracy. In addition, a comparative study of two fault diagnosis methods based on signal-to-image mapping is conducted. One representative model, VSI-DGCAPN, employs Vibration Signal Imaging (VSI) principles [41]; the other method is based on a Gram's angle field (GAF) image representation [32], which combines an improved fast kurtogram (IFK) with a nonlinear mode decomposition (NMD) to extract important features. Decomposition (NMD) to extract the physically significant feature components of the vibration signal and map them into a two-dimensional image using GAF, and finally, a convolutional neural network (CNN) to accomplish high-precision fault classification. The method is called GAF-CNN. As shown in Table 8, the methods proposed in this paper are superior to many advanced models mentioned above in terms of classification accuracy. Comprehensive experimental results and analysis show that the model in this paper exhibits excellent robustness and fault identification ability in

dealing with complex working conditions and multi-source signals, which verifies its wide applicability and superior performance in practical engineering applications.

5. Conclusion

This study proposes a TFRP feature map construction method based on EDWPD, effectively addressing the shortcomings of traditional recursive graphs (RP) in processing long-term signals and feature extraction richness. The EDWPD method adaptsively determines the number of wavelet packet decomposition layers, balancing the needs of signal denoising and feature retention. If the number of decomposition layers is insufficient, it will result in inadequate signal denoising, affecting the extraction of effective features; conversely, an excessive number of decomposition layers may reduce the signal-to-noise ratio due to multiple filter processing, potentially causing signal distortion and leading to the loss of useful information.

Compared to traditional RP methods, TFRP feature maps offer advantages such as shorter construction cycles and a more detailed representation of fault features, making them more suitable for fault diagnosis tasks. Building on this, EVD is further introduced to extract features. The extracted features are then input into the proposed PSF-CNN-BiLSTM model, which includes a feature extractor and a classifier module. The feature extractor provides extracted features to the classifier module. The dual-channel convolutional encoder structure progressively expands the receptive field and enhances feature abstraction capabilities. The outputs from the two channels are fused through feature fusion to obtain a joint multi-scale feature representation. The classifier module includes a BiLSTM neural network, which performs fault classification and diagnosis.

The proposed PSF-CNN-BiLSTM model was validated on two publicly available bearing fault datasets: CWRU and Paderborn. Experimental results show that on the 10-classification task of the CWRU dataset, the model achieves a diagnostic accuracy of 99.78 %; on the 7-classification task of the Paderborn dataset, the accuracy reaches 99.90 %. Compared with existing models, the proposed method achieves improved diagnostic performance. Additionally, feature visualization results based on T-SNE further validate the advantages of this method in feature separation and classification capabilities.

Future research will focus on the following directions: First, we will develop adaptive wavelet basis selection algorithms and dynamic recursive graph construction strategies based on deep learning to achieve dynamic optimization of feature representation. Second, we will study lightweight fusion architectures that significantly reduce computational complexity while maintaining diagnostic accuracy. Finally, a multi-modal intelligent diagnostic framework integrating multi-source information, such as vibration and temperature, will be constructed. Through cross-modal feature alignment and collaborative learning, the framework will enhance the model's applicability and robustness in complex industrial scenarios.

CRediT authorship contribution statement

Weize Dong: Writing – original draft. **Yulei Zhang:** Methodology. **Liwei Xu:** Conceptualization. **Daikong Jia:** Data curation. **Zheng'ang Shan:** Formal analysis. **Ningyuan Cui:** Resources. **Chunyue Gao:** Software. **Yang Liu:** Writing – review & editing, Conceptualization. **Hui Ma:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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